

CARMi, IL, USA

WMO: 722074

Lat: 38.089N

Lon: 88.123W

Elev: 388

StdP: 14.49

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 63840

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	9.1	13.9	0.4	5.7	11.0	5.1	7.2	15.6	27.9	42.1	25.6	36.6	6.4	320	0.459

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.6	93.7	75.7	91.3	75.3	89.8	75.0	80.2	88.0	78.4	87.1	77.1	86.0	8.1	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
78.1	148.0	85.0	76.0	137.8	83.1	74.5	130.9	81.6	43.9	88.0	42.2	87.4	40.8	86.3	85.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
23.5	19.9	17.7	1.7	97.7	4.0	3.3	-1.2	100.1	-3.6	102.0	-5.8	103.9	-8.7	106.3	
			DB												
			WB	0.7	82.2	3.7	2.2	-2.0	83.8	-4.1	85.0	-6.2	86.2	-8.9	87.8

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	56.7	33.3	36.4	45.9	57.5	67.3	76.2	76.6	75.5	69.7	57.6	45.4	37.4
	DBStd	17.92	11.06	11.08	10.08	9.18	8.20	5.28	4.60	5.15	6.56	9.26	9.31	9.56
	HDD50	1783	525	398	200	35	1	0	0	0	0	28	193	403
	HDD65	4449	982	800	593	254	76	2	0	2	29	269	588	854
	CDD50	4222	8	18	74	259	538	785	824	790	591	265	56	14
	CDD65	1412	0	0	3	28	148	337	359	327	169	40	1	0
	CDH74	13784	0	1	28	290	1353	3439	3407	3057	1747	451	11	0
	CDH80	5522	0	0	1	47	463	1506	1406	1266	702	130	1	0
Wind	WSAvg	7.1	9.2	8.8	9.3	9.0	7.3	6.0	4.4	3.9	4.5	6.7	7.6	8.1
Precipitation	PrecAvg	45.7	3.6	2.8	3.9	4.5	5.3	3.9	3.6	3.4	3.2	3.3	4.1	3.6
	PrecMax	56.1	7.6	5.9	8.9	12.0	10.6	8.1	7.4	9.3	6.4	8.5	8.5	7.0
	PrecMin	35.4	0.8	1.0	1.9	1.7	1.9	1.6	1.2	0.4	0.2	0.6	0.3	1.2
	PrecStd	6.5	1.5	1.3	1.8	2.6	2.7	1.7	1.8	2.4	1.9	2.1	1.9	1.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	63.8	70.5	77.7	85.6	91.5	98.8	97.1	96.9	94.9	89.6	74.9	66.2
		MCWB	57.6	60.3	63.0	65.9	72.3	72.5	76.1	76.1	72.9	70.1	62.4	61.3
	2%	DB	60.4	65.0	72.3	81.0	88.0	93.2	92.3	92.9	90.9	83.4	70.1	61.2
		MCWB	55.7	58.5	60.3	64.9	70.6	75.1	77.7	76.4	73.7	68.1	60.4	57.9
	5%	DB	54.8	59.3	67.7	77.4	85.4	90.9	90.2	90.1	88.0	79.4	65.2	56.5
		MCWB	51.1	53.7	57.9	63.0	69.5	74.1	77.3	76.2	73.0	65.8	57.3	52.7
	10%	DB	49.8	54.3	62.6	73.4	82.1	88.5	87.9	87.6	84.0	73.9	61.1	52.2
		MCWB	46.4	48.8	55.1	61.8	68.3	73.5	76.2	75.7	70.5	63.1	53.8	49.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	60.5	62.9	65.1	70.2	75.3	79.8	82.6	82.3	78.0	74.0	67.0	62.4
		MCDB	63.2	67.4	73.7	79.9	84.6	89.2	90.2	89.4	88.0	84.4	72.4	65.5
	2%	WB	56.4	59.2	62.3	67.8	73.4	78.1	80.7	80.0	76.4	71.3	62.7	58.5
		MCDB	59.2	63.9	70.2	77.2	83.6	88.6	88.2	87.7	86.7	79.2	67.1	61.0
	5%	WB	52.0	54.8	59.3	65.4	71.9	76.9	78.8	78.1	74.7	68.2	58.5	53.7
		MCDB	54.5	58.7	65.0	74.4	81.5	87.2	86.6	86.6	84.3	75.9	63.6	55.4
	10%	WB	45.8	49.0	56.0	63.4	70.1	75.4	77.2	76.6	72.6	65.4	54.7	48.8
		MCDB	48.9	53.2	62.1	70.4	79.0	84.6	85.2	84.8	80.6	72.1	60.2	51.6
Mean Daily Temperature Range	5% DB	MDBR	15.4	16.7	18.4	21.4	20.1	20.7	19.6	21.0	24.1	23.3	19.8	15.0
		MCDBR	21.7	24.6	25.2	25.7	23.7	23.5	22.0	24.4	28.0	28.3	25.9	20.6
		MCWBR	18.1	19.0	15.4	13.4	9.5	8.7	9.9	10.6	11.9	14.0	16.9	16.9
	5% WB	MCDBR	19.7	22.6	22.6	21.5	20.8	20.5	19.4	20.3	23.7	23.8	21.7	18.4
	MCWBR	18.2	18.9	15.7	13.1	9.5	9.0	10.3	10.4	11.2	12.6	17.2	17.0	
Clear-Sky Solar Irradiance	taub	0.314	0.327	0.366	0.403	0.437	0.445	0.472	0.475	0.413	0.361	0.326	0.311	
	taud	2.456	2.399	2.350	2.254	2.212	2.210	2.145	2.125	2.291	2.431	2.472	2.468	
	Ebn at Noon	275	285	283	278	269	266	258	254	264	270	268	268	
	Edh at Noon	26	31	36	43	45	45	48	48	38	30	26	24	
All-Sky Solar Radiation	RadAvg	631	862	1188	1570	1765	2025	1953	1787	1486	1094	741	541	
	RadStd	64	97	110	92	153	128	125	95	101	115	82	49	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (35 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page